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Original article

BIA-assessed cellular hydration and strength in healthy older adults

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SUMMARY

Objectives: The objective of this research was to evaluate the differential relationships between various BIA-estimated compartments of body water and strength, among healthy older adults, by age range and sex.

Design: Descriptive and correlational study.

Setting and participants: A total of 234 healthy older adults (72.59 ± 7.60 years) participated in the study. Participants were recruited from community centers around Portugal and Spain. Data were recorded during May 2023. This is a physically active sample: all the participants are physically active for a minimum of three days a week.

Methods: For the study of body composition the variables extracted from the Tanita were: fat mass (FM), fat free mass (FFM), visceral fat rating (VFR), total body water (TBW), extracellular water (ECW) and intracellular water (ICW). Handgrip Strength was used to evaluate upper body strength and the Chair Stand Test for lower body strength.

Results: Differences across the various data sets were recorded in the gender variable.

When looking into the gender variable, significant correlations were recorded for the body composition variables analyzed for all age groups in the upper body strength test, except for BMI ($r = .125-.878$) and Fat mass ($.056$ to $-.634$). For the lower body strength test, no significant correlations were recorded for the Fat-Free mass variables ($r = .164-.380$), as well as for the Visceral Fat Rating ($r = .082-.213$) and the body water variables (TBW, ECW, ICW).

Conclusions: The hydration parameters analyzed can be related to muscle performance for Handgrip in an independent older adult population with ages ranging from 60 to 90 years.

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1. Introduction

There are well-documented physiological changes that occur in the human body during the aging process, such as a decrease in fat free mass (FFM) - including lean body mass (LBM), bone mass (BM) and total body water (TBW) - and an increase in fat mass (FM) due to an increase in visceral fat mass (VFM) [1,2]. The aging process is also associated with a decrease in the desire to drink (reduced thirst), making older adults more vulnerable to dehydration [3]. Alterations in the body's hydration status have consequences for correct body function, with an increase in muscle injuries [4] and

reduced physical performance [5]. Other studies carried out on older adults suggest that chronic low-grade cellular dehydration has a relevant effect on muscle function [6,7]. The degree of cellular dehydration can have a great impact on the activities of daily living, physical activity, sport and even general health [8].

Dehydration in older adults is associated with a high risk of adverse health outcomes and death [9]. Dehydration contributes to many of the major causes of death and morbidity in older people. The Dehydration Council suggests that dehydration is a complex condition resulting in a reduction in TBW [10]. Decreased water intake or excessive water excretion can lead to dehydration. However, age-related dehydration does not seem to affect extracellular water (ECW) and intracellular water (ICW) equally. The increase in ECW relative to ICW with age has been described and put forward as a biomarker of muscle aging [11].

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Bioelectrical impedance analysis (BIA) is a valid, reliable, non-invasive, safe and portable method for estimating TBW, ICW and ECW [1]. BIA accurately measures water distribution (extracellular and intracellular water content) in the individual by assessing body composition [12]. In the research presented by Hetherington-Rauth et al., 2020 [12], the authors have indicated that BIA-derived hydration parameters are useful markers of muscle performance in all age groups.

Approximately 45 % of body weight in older adults is muscle and approximately 76 % of that muscle is water, indicating that muscle is the main reservoir of water in the body [13]. Therefore, it is relevant to investigate how body water and its compartments are related to muscle function. Because of the above, it was hypothesized that the degree of hydration in active older adults would influence the performance of strength in the upper and lower body. Therefore, the objective of this research was to evaluate the differential relationships of BIA-estimated body water compartments with strength among healthy older adults, according to age and sex.

2. Materials and methods

2.1. Subjects

This study is descriptive and correlational. A total of 234 healthy older adults (72.59760 years) participated in the study. Participants were recruited from community centers around Portugal and Spain, and data were recorded in May 2023. Participants were recruited through the European Erasmus + “IN COMMON SPORTS” project, which aims to organize adapted sports competitions based on the participants’ age specificities. This was a physically active sample, comprising participants who are physically active for a minimum of three days a week. The ethical standards of the Declaration of Helsinki were followed, and this study was approved by the Instituto Politécnico de Viana do Castelo Technical-Scientific Council (IPVC-ESDL180417). Exclusion criteria included the lack of ability to perform muscle performance testing due to health problems, mobility limitations, and/or difficulty understanding the instructions for the assessments, or failure to meet the prerequisites for BIA body composition assessment. Prerequisites for the BIA body composition assessment included no strenuous exertion having been done in the 12 h prior to the assessment, and the bladder having been emptied in the 30 min before assessment.

Further analyses divided the participants by gender and into three different age groups: group 1 (up to 70 years old), group 2 (70–80 years old), and group 3 (more than 80 years old). This division was made to see whether behavior patterns were maintained over time, thus verifying the effect of age across the tests covered.

2.2. Assessment

2.2.1. Anthropometric and body composition

A basic anthropometric assessment of weight and height was performed, and the body mass index (BMI) was calculated. Anthropometric measurements were performed following the protocols of the International Working Group of Kinanthropometry (ISAK) [14]. Weight was measured by means of a Tanita MC-780 (Amsterdam, Netherlands) multifrequency segmental body composition analyzer, with an accuracy to the nearest 100 g. For the study of body composition the variables extracted from the Tanita were: fat mass (FM), fat free mass (FFM), visceral fat rating (VFR), total body water (TBW), extracellular water (ECW) and intracellular water (ICW). This model requires the variables of sex, age and height to be entered, the latter being measured by means of a SECA (SECA, Germany) measuring rod with an accuracy of .1 cm. The

Tanita MC-780 takes about 20 s to collect the data once the indicated variables have been entered. The Body Mass Index (BMI) was calculated by dividing weight (kg) by height (m²). Visceral Fat Rating was estimated indirectly and the results were given as a specific rating with ranges from 0 to 59 (no units). Ratings from 1 to 12 indicate that the subject has a healthy level of visceral fat, while ratings from 13 to 59 indicate that the patient has an excessive level of visceral fat [15]. This newly developed BIA technology has been recently validated in adults of various physical activity levels [16].

2.2.2. Physical fitness. Strength

HandGrip Strength. Maximum grip strength was measured with a Jamar hand dynamometer (Sammons Preston Inc., Bolingbrook, IL, USA). The participants were asked to stand and hold the dynamometer with their arm parallel to their body, without squeezing it against their body. Both hands were evaluated in this test. The examiner ensured that the arm to be tested was held by the participant's side with their elbow at a 90° angle. The width of the handle was adjusted to the size of each participant's hand to make sure that the middle phalanx rested on the inner handle. Participants were allowed to perform one trial test. After this, three tests followed, with the best score being used for analysis. Muscle strength was expressed in kilograms (Kg).

Chair Stand Test. This test is another important test of functionality as it assesses the strength of lower body, which in turn determines activities of daily living (getting out of chair, climbing stairs, bed transfer) [17] for a 30 s period. Participants stand up from a seated position on a standard chair to a standing position, with full leg extension, and return, as many times as possible in the time period, with their arms folded across their chest. The number of completed cycles the individual can perform in 30 s is recorded. This test was then further undertaken, validated and normative scores published for American adults in the age groups of 60–94 [17,18].

2.3. Procedure

The data were recorded in three different locations, two in Portugal (Vilanova de Cerveira and Viana do Castelo) and one in Spain (Portonovo), during the month of May 2023. The evaluators were the same for the three collections, as was the equipment used. The data were collected over the course of a morning at each location. First, the anthropometric and body composition data were recorded, followed by a warm-up and the different physical tests, organized into stations.

2.4. Statistical analysis

Data are presented as mean $\pm$ standard deviation (SD). Data normality was checked by applying the Shapiro–Wilk test. Descriptive statistics, such as means and standard deviations, were calculated for gender and age groups. Partial correlations were used to analyze the degree of association between BIA (TBW, ECW, ICW, ICW:ECW ratio) and strength (Handgrip and Chair Stand Test) for each age group with a gender covariate. A correlation was made between Handgrip and total water for the whole sample, with the sex covariate. All statistical analyses were carried out using SPSS for Windows Version 27.0 (SPSS Inc., Chicago, IL, USA). P-values of $<.05$ were considered statistically significant.

3. Results

The participants’ characteristics are displayed in Table 1. Differences across the various data sets were recorded for the gender variable. For BMI, the sample remained within normal values. For

Table 1
Descriptive analysis of the body composition and strength of healthy older adults organized by sex and age groups.

	All (n=234)			Male (n=64)			Female (n=170)		
	<69 n=100	70 a 80 n=96	>81 n=38	<69 n=24	70 a 80 n=23	>81 n=17	<69 n=76	70 a 80 n=73	>81 n=21
Age, years	66.06±4.04	74.94±2.70	87.00±3.85	66.13±2.98	75.70±2.65	88.71±4.35	66.04±4.33	74.60±2.68	86.43±3.60
Weight, kg	71.10±12.72	68.23±10.81	65.52±11.70	79.58±11.82	76.27±8.69	72.44±6.93	68.42±11.85	64.59±9.68	63.21±12.17
Height, cm	158.31±8.51	155.55±7.12	152.55±8.97	169.45±6.60	162.35±5.39	163.71±7.79	154.80±5.48	152.47±5.49	148.83±5.69
Body Composition									
BMI, kg/m ²	28.37±4.52	28.15±3.73	28.07±3.84	27.77±4.12	28.92±2.82	27.03±2.08	28.55±4.65	27.80±4.05	28.41±4.25
Fat Mass, kg	22.67±7.70	20.67±6.12	21.37±6.46	19.00±6.77	17.55±4.98	17.00±3.72	23.83±7.66	22.08±6.09	22.83±6.59
Fat-Free mass, kg	48.43±8.75	47.56±9.13	44.15±9.28	60.58±6.64	58.73±5.46	55.44±6.43	44.59±5.03	42.51±5.02	40.39±6.65
Visceral Fat Rating	10.39±3.17	11.63±3.18	13.32±2.97	13.79±3.40	15.21±2.38	17.14±2.54	9.32±2.20	10.01±1.92	12.05±1.77
TBW, kg	34.06±6.15	33.53±6.80	30.51±6.39	42.38±5.02	41.80±4.53	37.84±4.98	31.43±3.61	29.80±3.58	28.07±4.75
ECW, kg	15.48±2.44	15.34±2.55	14.75±2.57	18.45±1.79	18.29±1.45	17.53±1.43	14.55±1.78	14.00±1.65	13.83±2.18
ICW, kg	18.57±3.81	18.20±4.34	15.76±3.90	23.93±3.31	23.51±3.16	20.31±3.63	16.88±1.93	15.80±2.06	14.24±2.62
ICW:ECW ratio	1.19±0.09	1.18±0.11	1.06±0.09	1.29±0.08	1.28±0.09	1.15±0.13	1.16±0.06	1.13±0.07	1.03±0.05
Bone Mass, kg	2.45±0.42	2.41±0.44	2.25±0.45	3.03±0.31	2.95±0.25	2.80±0.31	2.27±0.25	2.17±0.25	2.06±0.33
Physical Fitness (Strength)									
Handgrip, kg	27.35±7.32	27.01±6.15	26.75±10.81	41.17±8.47	34.20±4.57	33.00±10.58	25.54±4.90	24.25±4.11	20.00±4.01
Chair Stand Test, n	15.38±3.38	17.04±5.23	16.50±2.52	18.67±4.59	19.00±5.69	16.67±3.06	25.11±5.56	24.27±4.52	20.00±5.15

BMI: Body mass index; TBW: total body water; ECW: extracellular water; ICW: intracellular water.

fat mass, men presented values below 20 kg, while women presented values between 22 and 24 kg. Regarding visceral fat, women presented healthy values, while men exceeded these values in the lower limit.

Regarding the sex variable, significant correlations were recorded for the body composition variables analyzed across all age groups in the upper body strength test, except for BMI and Fat mass. For the lower body strength test, no significant correlations were recorded for the Fat-Free mass variables, such as for the Visceral Fat Rating or for the body water variables (TBW, ECW, ICW). For the 80+ sample, inverse relationships were recorded for the strength and body composition parameters (except Visceral Fat Rating) (see Figs. 1 and 2 and Table 2).

The level of correlation of maximum handgrip strength (Handgrip, Kg) with total body water (TBW) presents a value of $r = .450^{**}$. This indicates a positive and significant correlation with an average correlation level.

4. Discussion

The present study analyzed the relationship between the degree of hydration and upper and lower body strength performance in active older people. The results show that the degree of hydration influences upper body strength results across all the age ranges analyzed. These correlations were not found between hydration variables and lower body strength, except in older adults aged 70–80 years, for the ICW:ECW ratio.

The level of body water, as well as its location in the different compartments, can predict the muscular performance of the upper body in older adults, this prediction being unaffected by which age group is evaluated, with the ICW:ECW ratio being the variable that presented the lowest value in the correlation. The results presented by the study of Hetherington-Rauth et al. [12] conclude, in a more detailed manner, that for older adults (over 64 years), the ICW estimated by BIA seems to be the best predictor of muscle performance. Previous research pointed out that ICW and ICW:ECW ratio are important indicators of muscle performance [7,12,19,20,21]. Kuchnia et al. [19], revealed that ICW also reflects muscle contractile tissue. With age, there is a steady loss of muscle fibers and a corresponding accumulation of fat within the muscle tissue, which corresponds to a decrease in ICW [21].

The greater heterogeneity of ICW with age probably explains the stronger relationships observed between ICW and muscle performance metrics in adults and older adults, as compared to those of children and adolescents [12]. In this study the ICW presented a positive and significant correlation in all age ranges analyzed for Handgrip strength. These results are in line with the study by Hetherington-Rauth et al. [12], but not with the research conducted by Serra-Prat et al. [7]. The variables ECW and ICW:ECW ratio also presented positive and significant correlations across all the age ranges analyzed for hand grip strength. This behavior differs from that found in several other studies [7,12,21,22] of older adults, where researchers indicated that the ECW constitutes the interstitial fluid and blood plasma in the extracellular space between muscle fibers and, therefore, that increased extracellular water in skeletal muscle tissue is an important contributor towards decreased muscle quality during aging [11,23].

In Ritz's study [24], he compared the ECW values of healthy and unwell older adults and the values were higher in the healthy group. This should be taken into account in the interpretation of these results, since in this study we evaluated a sample of physically active independent older adults, and the relationship of hydration variables on upper body strength was analyzed. There are

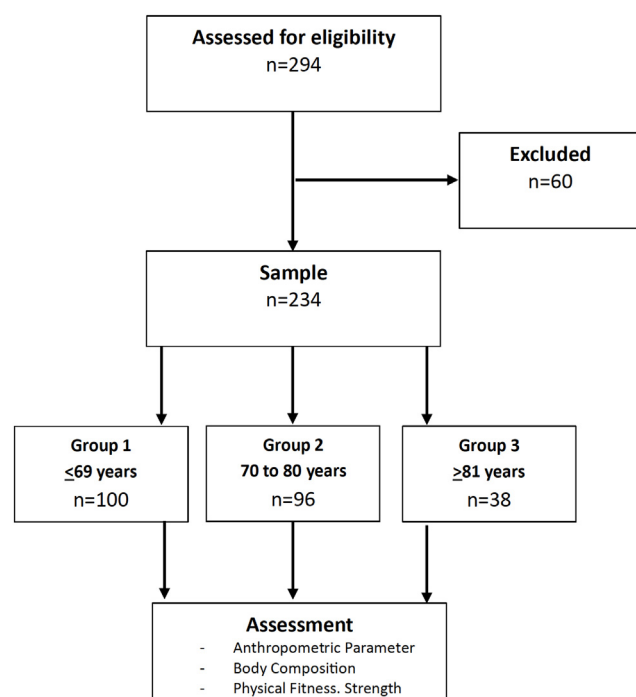


Fig. 1. Study flow chart.

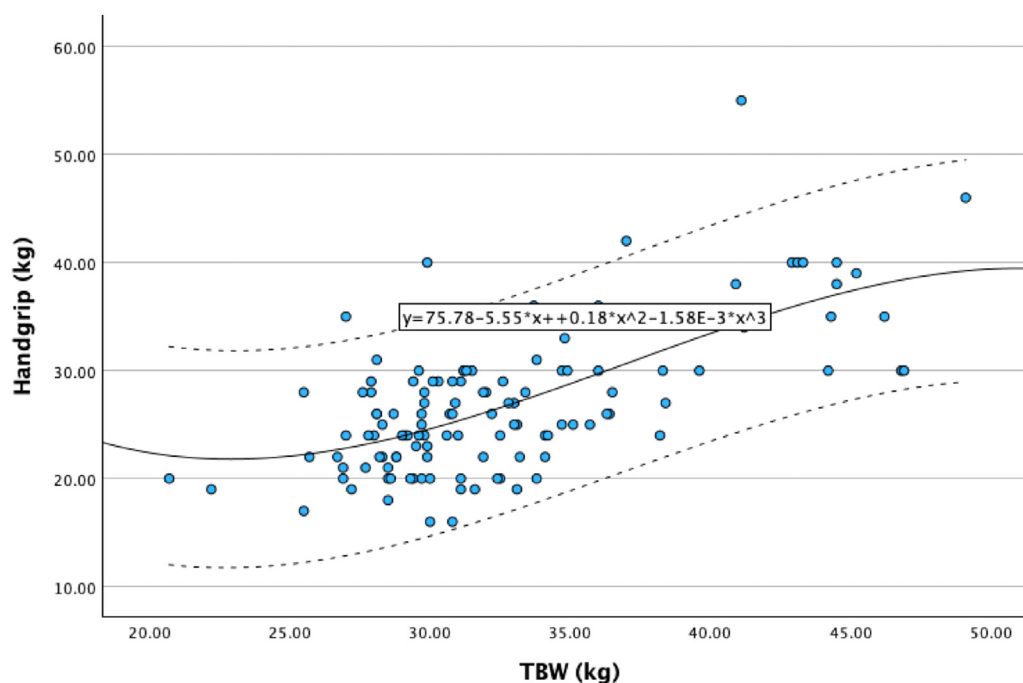


Fig. 2. Correlations between handgrip and total water (TBW).

Table 2

Correlational analysis between body composition indicators and strength level by age groups.

	Strength			Lower Limb		
	Upper Limb			Chair Stand Test		
	HandGrip					
	<69 n = 100	70 to 80 n = 96	>81 n = 38	<69 n = 100	70 to 80 n = 96	>81 n = 38
Body Composition						
BMI, kg/m ²	.125	.130	.878	-0.058	-0.063	.687
Fat Mass, kg	.056	-0.198	-0.634	-0.161	-0.244*	-0.239
Fat-Free mass, kg	.654**	.715**	.712**	.164	.173	-0.380
Visceral Fat Rating	.497**	.572**	.620**	.208	.082	.213
TBW, kg	.647**	.697**	.753**	.163	.193	-0.344
ECW, kg	.587**	.673**	.686**	.112	.135	-0.306
ICW, kg	.672**	.695**	.632**	.195	.224	-0.367
ICW: ECW ratio	.468**	.505**	.522**	.216	.288*	-0.389
Bone Mass, kg	.654**	.703**	.706**	.167	.176	-0.385

BMI: Body mass index; TBW: total body water; ECW: extracellular water; ICW: intracellular water. * sig.<.05, ** sig.<.001.

very few published studies reporting the effect of ECW on muscle function and frailty in elderly individuals, which calls for further large, prospective studies.

With respect to the strength levels obtained for the lower body (Chair Stand Test) and the relationship of these to body hydration, we should point out that these data do not agree with some other studies which suggest that the ICW is related to strength levels and muscle mass [7,12,24,25], since only the ICW:ECW ratio for adults between 70 and 80 years of age demonstrated a significant correlation. It should be noted that the aforementioned studies analyzed jumping to measure lower body strength, while in this research the sit-to-stand test was used, so the method for quantifying strength differs. It could be concluded that the test used in the current study is inadequate for the analysis of how the level of body hydration affects the behavior of an active population, but in studies carried out with older adults, the Chair Stand Test is one of the most widely used and standardized. Another aspect to highlight in the results of this study is the inverse character of the variables analyzed for body composition in the sample made up of adults over 80 years of age. This may suggest that the increase in the variables analyzed has an

inverse behavior with strength, which would be in line with the previously mentioned studies [7,12,24,25]. The low sample number of this group should be stressed, as it meant that there was inadequate statistical power for analysis.

The results obtained show the importance of hydration as a conditioning factor for upper body strength, suggesting that older adults should have an intake adapted to their level of physical activity, so as not to reduce their muscular strength, although future research should focus on determining the optimal amount of fluids to be ingested by this group. On the other hand, and due to the results obtained when comparing body water and lower body strength level, it would be good to replicate the present investigation to determine the cause of the differences in behavior between lower and upper body strength levels with respect to body water.

5. Limitations

The test used to quantify lower body strength levels (Chair Stand Test) is one of the most widely used to assess functional capacity (Strength) for the independent older adult, but in view of the

results, it is perhaps not the most appropriate for comparing its values with the hydration level of active older adults. It would be appropriate to test the behavior of the Chair Stand test with respect to the level of body water in sedentary older adults. Although the sample is significant, when segmented into age groups, in the population over 80 the number of subjects should be increased. For the analysis we have used the sex covariance for the analysis; it would also be appropriate to approach the study segmented by sex.

6. Conclusions

The hydration parameters analyzed can be related to muscular performance for Handgrip in an independent older adult population with ages ranging from 60 to 90 years. No relationship was found between the degree of dehydration and low muscle performance for the Chair Stand Test in physically active independent older adults.

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Institutional review board statement

This study was conducted according to the guidelines of the Declaration of Helsinki and approved by the Ethics Committee of the Polytechnic Institute of Viana do Castelo (IPVC-ESDL180417).

Informed consent statement

Informed consent was obtained from all subjects involved in this study.

Data availability statement

The data presented in this study are available on request from the corresponding author.

Declaration of competing interest

The authors declare no conflicts of interest.

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